THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS TWO-CHAPTER TEST SET B

Chapters 1 & 2: **A Square and a Cube, Power Play** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The value of 9^3 is: 1

(a) 729 **(b)** 243 **(c)** 738 **(d)** 720

2. $\sqrt[3]{729}$ is: 1

(a) 11 **(b)** 9 **(c)** 10 **(d)** 8

3. Which of the following is a perfect cube? 1

(a) 130 **(b)** 125 **(c)** 200 **(d)** 150

4. The number 1729 is famous as the: 1

(a) largest perfect cube
(b) smallest perfect square
(c) smallest number expressible as the sum of two cubes in two ways
(d) sum of first 10 odd numbers

5. The smallest number by which 72 must be multiplied to get a perfect square is: 1

(a) 3 **(b)** 2 **(c)** 6 **(d)** 4

6. $\left(\frac{2}{5}\right)^3$ equals: 1

(a) $\frac{6}{15}$ **(b)** $\frac{8}{15}$ **(c)** $\frac{8}{125}$ **(d)** $\frac{2}{125}$

7. The number 6300000 in standard (scientific) form is: 1

(a) 6.3×10^5 **(b)** 6.3×10^7 **(c)** 63×10^5 **(d)** 6.3×10^6

8. One crore is equal to: 1

(a) 10^7 **(b)** 10^5 **(c)** 10^8 **(d)** 10^6

9. 7.5×10^4 written in usual form is: 1

(a) 7500 **(b)** 0.00075 **(c)** 750000 **(d)** 75000

10. If $2^x = 64$ then x is: 1

(a) 8 **(b)** 5 **(c)** 6 **(d)** 7

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. What is a Pythagorean triplet? Write the triplet that contains 8. 2

12. Why is 1729 called the Hardy–Ramanujan number? 2

13. Write the number 3,500,000 in scientific notation. 2

14. What is the speed of light, 300000000 m/s, in scientific notation? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Observe the pattern: $1^2 = 1$, $11^2 = 121$, $111^2 = 12321$. Predict 1111^2 and 11111^2. 3

16. Simplify $2^{-3} + 3^{-2}$ as a single fraction. 3

17. Find the square of 78 using $(a+b)^2$ with $a = 80$, $b = -2$. 3

OR

Is 1372 a perfect cube? If not, by which smallest number should it be multiplied to make it one? 3

18. Find the value of $\left(2^3\right)^2 \times 2^{-4} \div 2^0$. 3

OR

The mass of the Earth is 5.97×10^{24} kg and of the Moon is 7.35×10^{22} kg. Which is greater and about how many times? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. A rectangular field of area 1176 m² has length and breadth in the ratio $3 : 2$. Find its sides and the cost of fencing at ₹20 per metre. 5

OR

(i) Show that $\sqrt{a} \times \sqrt{b} = \sqrt{ab}$ using $a = 36$, $b = 49$. 5
(ii) Find $\sqrt{2916}$ by factorisation.

20. (i) The diameter of a hydrogen atom is about 10^{-10} m and of a pollen grain about 10^{-5} m. How many times larger is the pollen grain? 5
(ii) Express a red blood cell of 7×10^{-6} m in mm.

OR

(i) Explain how to compare two numbers written in scientific notation. 5
(ii) Compare 4.5×10^7 and 9.1×10^6.

END OF QUESTION PAPER — Total: 40 Marks

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